

Generative Market Simulation

Update: Mechanism · Market Design · Engine Update

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① Designing for Emergence

② Mechanism

③ This Week

Two concurrent studies from early 2025 reveal contrasting LLM market behaviors, highlighting that outcomes are primarily driven by market structure.

Too efficient

LLM Agents Do Not Replicate Human Market Traders

- Homogeneous agents, simple environment
- Price $\approx$ fundamental value
- Low strategy variance, no bubbles

Emergence works

TwinMarket, NeurIPS 2025

- Heterogeneous agents + social feed
- Bubbles, crashes, herding emerge
- Human-like stylized facts

Takeaway

Studying strategy requires environments with **rich structure**.

To observe true strategy emergence, the environment need satisfy two conditions at once:

1. Rich enough — provide exploitable structure

No structure $\Rightarrow$ no strategy.

- **Information:** ambiguous public news that needs interpreting
- **Market design:** correlated markets (A, B, $A \vee B$) for arbitrage
- **Action space:** open actions — transfers, creating markets

2. Clean enough — minimize artificial intervention

We design the environment, never the strategy.

- **No pre-assigned roles:** we never dictate who errs or inject bias
- **Organic only:** any anomaly must arise from the LLMs themselves

Rich enough for strategies to emerge — clean enough that whatever emerges is genuinely the agents' own.

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Each round: the same fixed 4-step loop



Step	What it does	Example (R1 · COIN-A)
READ	market state + news	get_news → 61% (± 4); book empty
COMMIT VIEW	belief + plan	commit COIN-A = 0.61
TRADE	place/cancel limit/market orders (<i>optional</i>)	bid YES@55 & @50 (under 61)
FINISH	finish signal + this round's reflection	"books empty; bid below signal"

The agent only picks the price it's willing to pay — it never sets the market price.

How do prices change?

The price remains fixed at the last trade until a new transaction occurs. Because traders order based on their beliefs, the price moves only when a shift in belief pushes a new order across the spread.

Worked example

Book: best bid **58¢** / best ask **62¢** $\Rightarrow$ market price $\approx$ **60¢**

Your belief	Best bid	Crosses @ 62?	Last trade = price
60%	60¢	 no	60¢ (unchanged)
63%	62¢	 yes	62¢ (price rises)

Key point

Price rose because the marginal buyer's bid crossed the ask — buying *below* 62¢ never moves it.

The role: Liquidity provision

Market makers act as continuous, two-sided counterparties. By quoting both bid and ask prices, they provide essential liquidity and earn the bid-ask spread.

Our setup

Instead of hard-coding bots, we implement Polymarket-style incentives. We allow market-making to emerge as an incentivized, optional role for agents.

*Polymarket mechanics: Liquidity Rewards, Maker Rebates

The question: Under these incentives, do LLM agents independently adopt market-making?

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- **Open-ended agent actions:** `transfer`, `create_account`, `create_market` — wired into LLM tools, the CLI/console, and a web panel.
- **Advanced order types** with Polymarket-aligned defaults (below).
- **Web scenario builder + templates:** a browser form generates a Config, saved as reusable YAML, runnable in-browser or via CLI.

Button	Default	Behavior
Market	FAK	fill what's available, cancel the rest
Limit	GTC	rest on the book until filled / cancelled
Advanced	FOK / GTD	all-or-none / expires

Polymarket has no native market order — the “market” button is FAK; the SDK default (no type specified) is GTC.

Thanks!

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